# Skills Summary

## Expressions: Distribute, Combine, Factor

*Use this reference while completing your worksheet or when studying.*

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### 1. Distributing over a sum (distribute-over-sum)

**Rule:** $a(b+c) = ab + ac$ and $a(b-c) = ab - ac$

**Why:** $a(b+c)$ is the area of a rectangle of height $a$ whose base is cut into $b$ and $c$. Cutting the base cuts the area into $ab$ and $ac$. The outside factor multiplies **every** term inside — and each term keeps the sign in front of it.

**Example:** Expand $6(x+5)$.
**Step 1.** Two terms sit inside one set of parentheses, so this is a distribution: multiply the outside factor by each term.
**Step 2.** $6 \cdot x = 6x$ and $6 \cdot 5 = 30$, so $6(x+5) = \mathbf{6x + 30}$

**Try it:** Expand $3(4x+7)$.

check: $\mathbf{12x + 21}$

### 2. Combining like terms (combine-like-terms)

**Like terms** have exactly the same variable part: $5x$ and $3x$ are like; $5x$ and $5$ are not.
**Rule:** add or subtract the coefficients and keep the variable part: $5x + 3x = 8x$.

Every term travels with the sign in front of it, so $9 - 4$ is a subtraction of like number terms, not a stray minus.

**Example:** Simplify $5x + 9 + 3x - 4$.
**Step 1.** There are no parentheses, only terms of two different kinds, so this is pure sorting: group like with like.
**Step 2.** $(5x + 3x) + (9 - 4) = \mathbf{8x + 5}$

**Try it:** Simplify $7x + 12 - 2x - 5$.

check: $\mathbf{5x + 7}$

## 3. Distribute, then combine (distribute-then-combine)

**Order matters:** you can only combine terms once nothing is trapped inside parentheses.

**Step 1** distribute every product. **Step 2** collect like terms.

A term *outside* the parentheses (like the $5x$ in $2(x+6)+5x$) is never multiplied by the outside factor — it only joins in at step 2.

**Example:** Simplify $2(x+6)+5x$.
**Step 1.** Part of this expression is locked inside parentheses, so distribute first and only then look for like terms.
**Step 2.** $2(x+6) = 2x+12$, then $2x+12+5x = (2x+5x)+12 = \mathbf{7x+12}$

**Try it:** Simplify $4(3x-1)-2x$.

check: $\mathbf{10x - 4}$

## 4. Factoring out a common factor (factor-out-common-factor)

**Rule (distribution backwards):** $ab + ac = a(b+c)$

Find the **greatest** common factor of the terms, write it outside, and divide each term by it to get what stays inside. Check by distributing: you must land back on the expression you started with.

**Example:** Factor $8x+20$.
**Step 1.** Both terms share a whole-number factor, so run the area model backwards: the shared factor is the height.
**Step 2.** GCF of 8 and 20 is 4; $8x \div 4 = 2x$ and $20 \div 4 = 5$, so $8x+20 = \mathbf{4(2x+5)}$

**Try it:** Factor $9x-24$.

check: $\mathbf{3(3x - 8)}$

**(!) Watch out:** $3(2x-5) \neq 6x-5$. The outside factor must reach the second term too: $6x-15$. Test any doubtful step by substituting a number — equivalent expressions agree for every value.